MILITARY SPECIFICATION

DEXAMETHASONE SODIUM PHOSPHATE AEROSOL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers Dexamethasone Sodium Phosphate Aerosol (See 6.2).

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

PPP-C-186	- Containers, Packaging and Packing for Drugs, Chemicals, and Pharmaceuticals (See 6.3).

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Hq, Defense Personnel Support Center, ATTN: Directorate of Medical Materiel, DPSC-ATT, 2800 South 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

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(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

U. S. DEPARTMENT OF HEALTH, EDUCATION, AND WELFARE,
FOOD AND DRUG ADMINISTRATION

Federal Food, Drug, and Cosmetic Act and Regulations
Promulgated Thereunder.

(Application for copies should be addressed to the Food and Drug Administration, Washington, DC 20204.)

U. S. PHARMCOPEIAL CONVENTION, INC.

The United States Pharmacopeia.

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

National Formulary.

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

3. REQUIREMENTS

3.1 Material. Shall be Dexamethasone Sodium Phosphate Aerosol, an alcoholic suspension of Dexamethasone Sodium Phosphate in an inert propellant suitable for intranasal administration. The valve shall deliver a measured dose of approximately 0.1 mg Dexamethasone Phosphate with each depression of the valve. Shall contain 2.0 percent Alcohol within the designated assay limits. The inert propellant shall be a mixture of dichlorodifluoromethane and dichlorotetrafluorethane.

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3.1.1 Assay. Shall assay to contain not less than 1.7 percent and not more than 2.3 percent Alcohol, when determined by a suitable, accurate, reproducible gas-liquid chromatographic method. Each 10 sprays shall contain not less than 0.80 mg and not more than 1.20 mg Dexamethasone Phosphate based upon the weight of ten sprays and the mg of Dexamethasone Phosphate per gram of the spray found in the assay when tested as specified in 4.7.1.

3.1.2 Aerosol delivery. Shall deliver a minimum of 170 measured sprays when tested in accordance with 4.7.2. The aerosol shall deliver not less than 12.6 grams per container and shall contain not more than 1.4 grams of nonsprayable contents when tested as specified in 4.7.2.

3.1.3 Uniformity of delivery. Shall be as specified in 4.7.1.

3.1.4 Leakage. The average leakage rate shall be within the limits specified in the NF when tested by the NF Leak Test for Aerosol. The plastic protective coating shall be removed from the sides (below crimp) and bottom of the container prior to performing the test, if a plastic protective coating is supplied.

3.2 Ingredients.

3.2.1 Dexamethasone Sodium Phosphate and Alcohol. The Dexamethasone Sodium Phosphate and Alcohol used in the manufacture of the aerosol shall be in accordance with the tests, standards, and requirements of the USP including any supplements or revisions thereto.

3.2.2 Dichlorodifluoromethane and dichlorotetrafluoroethane. The dichlorodifluoromethane and dichlorotetrafluoroethane used in the manufacture of the aerosol shall comply with the tests, standards and requirements of the NF, including any supplements or revisions thereto,

3.2.3 Other ingredients. All other ingredients entering into the finished aerosol shall be of USP or NF quality, or, if not included in either of these compendia, the ingredients shall be of the highest pharmaceutical grade.

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3.3 Functionability at environmental temperatures. The immediate containers shall function properly and shall comply with the requirements of 3.1.2 for measured dose at an environmental temperature of +5.0° C. to +37.8° C.

3.4 Expiration dating period. The expiration dating period shall be 36 months.

3.4.1 Delivery. Not less than 33 months of the expiration dating period shall remain at time of delivery to the Government.

3.5 Workmanship. Workmanship shall be first class throughout. The component parts of the completed aerosol assembly shall be free from flaws or other defects which detract from their appearance or may impair the function of the assembly.

4. QUALITY ASSURANCE PROVISIONS

4.1 Supplier responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and, (i) as to any expiration dated item, for the expiration dating period specified by the contractor for such item, or such longer period as may be required by regulation of any federal agency; or (ii) as to non-dated items, for not less than 3 years after delivery of the items to the Government.

4.1.2 No company supplying any ingredient(s) to the contractor will be considered an acceptable facility for the performance of any inspection requirements specified herein.

4.2 Lot. For purposes of this specification, a lot, batch, or control is that single, uniform, and homogeneous quantity of aerosol produced from one formulation, subjected to the same compounding and manufacturing operation, and filled into final containers.

4.3 Sampling. Sampling shall be conducted in accordance with MIL-STD-105.

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TABLE I. Sampling

For Visual examination	Inspection level	AQL (percent defective)	Unit of product
Major	S-2	1.0	Unit package
Minor	II	2.5	Unit package

4.3.1 End item testing. Noncompliance with any of the following is cause for rejection.

TABLE II. Criteria for end item testing.

Characteristic	Requirement paragraph	Inspection level 1/	Minimum sample
Assay	3.1.1	S-2	5
Aerosol delivery	3.1.2	S-2	5
Uniformity of delivery	3.1.3	S-2	5
Leakage	3.1.4	NF	requirements

1/ Where possible, the sample shall be used for two or more tests.

4.4 Examination. The finished aerosol shall be examined to determine compliance with all requirements of this specification. Nonconformance will be permitted to the extent indicated in 4.3.

4.4.1 Classification of defects. Examination will be conducted in accordance with the following classification of defects:

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TABLE III. Classification of defects

Categories	Defects 1/
Major	
101	Valve does not function properly (continuous or no spray).
102	Valve item is loose in adapter.
103	Adapter not free of cracks or breaks.
Minor	
201	Exterior of immediate container is not free from contents and foreign matter.
202	Label not free from defacing marks and tears.
203	Adapter not clean.

1/ Inspection is not restricted to the classified possible defects listed above.

4.5 Packaging inspection. The inspection of the intermediate package, packing and marking for shipment and storage shall be in accordance with quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

4.6 Labeling inspection. Sampling shall be in accordance with MIL-STD-105 and the unit of product shall be the unit of issue. The inspection lot shall be the number of units of issue of each batch, lot, control number, etc., offered for inspection at one time. The inspection level for labeling shall be S-2 and the acceptance number shall be zero.

4.7 Tests.

4.7.1 Assay of Dexamethasone Phosphate.

Apparatus and Reagents:

1. Dry ice-acetone bath.

2. Sampling needle for Aerosol (see NF under Aerosols).

3. Porter-Silber Reagent. Carefully add 310 ml of concentrated sulfuric acid to 190 ml of purified water. To 100 ml of this solution, add 50 ml of isopropyl alcohol and 65 mg of phenylhydrazine hydrochloride.

4. Standard. Accurately weigh approximately 120 mg of USP Dexamethasone Sodium Phosphate Reference Standard into a 100-ml volumetric flask. Dilute to volume with purified water. Dilute 10 ml of this solution to 100 ml with purified water.

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Procedure:

Add 15 ml of methylene chloride to each of five 50-ml centrifuge tubes and place in a dry ice-acetone bath. Accurately weigh the container to be assayed and record this on the "initial weight." Spray 20 sprays into the air and reweigh the container. Remove one of the centrifuge tubes from the bath and, using the sampling needle, spray 10 sprays into the tube. Shake before each spray. Reweigh the container to determine the sample weight for tube No. 1. Pipet exactly 10 ml of 0.1N H_2SO_4 through the sampling needle into the centrifuge tube. Shake cautiously; allow the phases to separate and come to room temperature.

Repeat the above for the four remaining tubes; i.e., spray 20 sprays into the air, weigh, spray 10 sprays into the tube, pipet 10 ml of 0.1N H_2SO_4 into the tube, shake, allow the phases to separate and come to room temperature. Retain the container for delivery testing (4.7.2).

Pipet 1.0 ml of the aqueous layer from each tube into spearate glass-stoppered tubes. Pipet 1.0 ml of the Standard and 1.0 ml of purified water into separate glass-stoppered tubes. Add exactly 5.0 ml of the Porter-Silber reagent to each tube and shake. Heat the seven tubes in a water bath at 70°C. for 1-1/2 hours. Remove the tubes from the water bath and cool. Determine the absorbances of the samples and the standard at 410 nm in a 1cm-cell using a suitable spectrophotometer against a reagent blank. Calculate the Dexamethasone Phosphate per gram for each assay, as follows:

$$\frac{Au}{As} \quad C \times \frac{10}{W} \quad \times P = \text{mg Dexamethasone Phosphate per Gram}$$

where Au = Absorbance of the sample
As = Absorbance of the standard
C = concentration of standard in mg Dexamethasone Sodium Phosphate per ml
W = weight of sample (10 sprays) in grams
P = purity of standard expressed in % Dexamethasone Phosphate

The average of the five assays shall be not less than 1.29 mg per gram and not more than 1.57 mg per gram. No individual assay shall vary by more than 20 percent from the average of the five assays. Not more than one assay shall vary by more than 10 percent from the average of five assays.

As an alternate, the assay for Dexamethasone Phosphate may be determined by an appropriate, accurate and reproducible method. The assay method shall be submitted with the bid or proposal and shall be subject to approval and acceptance by the Government. If the method is approved and accepted by the Government, such method shall be used for the assay of Dexamethasone Phosphate. No further submission of the method is required, provided the contractor states that the product and assay method have not been modified in any way.

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4.7.2 Delivery. Using the container used for the assay (150 sprays removed), continue spraying, weighing the container every five sprays until the cumulative weight of five sprays is less than 315 mg or until the container is exhausted. At this point, not less than a total of 170 sprays shall have been expelled. Weigh the container. Determine the weight of sprayable contents as follows:

Grams sprayable contents = Initial Weight - Final Weight.

Chill the container in the dry ice-acetone bath. Carefully remove the seal and valve and discard the remaining contents. Rinse the container with alcohol and acetone. Place the empty container, valve, and seal in an oven at 105° C. for 2 hours, cool, and reweigh. Calculate the weight of nonsprayable contents as follows:

Grams nonsprayable contents = Final Weight - Container Weight.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with all applicable paragraphs of PPP-C-186 (see 6.3) and as specified herein.

5.2 Packaging. Packaging shall be level A or C, as specified (see 6.1).

5.2.1 Unit of issue. One package containing one vial equipped with a metering valve and a nasal adapter, as specified, constitutes one unit of issue.

5.2.2 Level A.

5.2.2.1 Immediate container. The immediate container shall be a commercially available vial equipped with a metering valve and a nasal adapter considered acceptable by the Food and Drug Administration (FDA) for the product contained therein and shall be of such compatibility that neither the contents nor the container are altered in any way by each other.

5.2.2.2 Unit package. Each unit shall be packaged as specified in 5.2.5 of PPP-C-186.

5.2.2.3 Intermediate package. Twelve unit packages shall be packaged as specified in 5.2.7.1 of PPP-C-186.

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5.2.3 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used which will afford the degree of protection required for shipment and use of the product for its intended use.

5.3 Packing. Packing shall be level A, B or C, as specified (See 6.1).

5.3.1 Level A. Twelve intermediate packages (144 units) shall be packed in an exterior container as specified in 5.4.4 of PPP-C-186.

5.3.2 Level B. Twelve intermediate packages (144 units) shall be packed in an exterior container as specified in 5.4.3.1.1 of PPP-C-186.

5.3.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.4 Marking.

5.4.1 Labeling. Labeling shall be in accordance with the requirements of the Federal Food, Drug and Cosmetic Act, and shall include the information required below.

5.4.1.1 Immediate container. Each immediate container label shall bear the following information:

(a) label information in accordance with commercial practice.

(b) the expiration date.

5.4.1.2 Unit package. Each unit package shall bear the following information:

(a) label information in accordance with commercial practice.
(b) the national stock number.

NOTES:

(1) The national stock number shall not be preceded by any prefix.

(2) Strip labeling may be used.

(3) Label adhesion test shall not apply.

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5.4.2 Intermediate package. Intermediate package shall be marked as specified in 5.5.3 of PPP-C-186. Type I Shelf life markings as specified in MIL-STD-129 shall be shown. Marking shall include the legend: "STORE AT CONTROLLED ROOM TEMPERATURE 15° - 30°C. (59° - 86°F)".

5.4.3 Exterior container. Exterior container shall be marked as specified in 5.5.4 of PPP-C-186. Type I shelf life marking as specified in MIL-STD-129 shall be shown. Marking shall include the legend: "STORE AT CONTROLLED ROOM TEMPERATURE 15° - 30°C. (59° - 86°F.)".

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) the National Stock Number.
(c) Selection of applicable levels of packaging and packing (See 5.2 and 5.3).

6.2 This specification covers the following item(s):

National Stock No.	Item Identification
6505-00-885-6302	DEXAMETHASONE SODIUM PHOSPHATE AEROSOL, 12.6 Gram, 170 Metered Doses

6.3 All references to PPP-C-186 refer to Interim Federal Specification PPP-C-00186B which is mandatory for use until superseded by a revision.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6505-1513

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE

MIL-D-37370 DEXAMETHASONE SODIUM PHOSPHATE AEROSOL

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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